MILITARY SPECIFICATION

SPRAYER AND DUSTER UNITS, INSECTICIDE, MIST-DUST BLOWER TYPE: GED, 30 to 100 GAL/H LIQUID, 30 to 300 LB/H DUST (COMMERCIAL)

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a gasoline-engine-driven insecticide sprayer and cluster unit of the mist-dust type, for outdoor distribution of chemicals by means of a high velocity air stream.

1.2 Classification. The sprayer and duster unit shall be one of the following types, as specified (see 6.2):

Type A - Trailer mounted.
Type B - Skid mounted.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2), the following specifications, standards, and handbooks of the issue listed in-that issue of-the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-V-173 - Varnish, Moisture and Fungus Resistant (for Treatment of Communications, Electronic and Associated Equipment).
MIL-S-12651 - Sprayers, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

2.1.2 Other -Government-documents. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations.
Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Code of Federal Regulations, Title 40, Part 86 - Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines: Certification and Test Procedures.

(Application for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as Department of Defense adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

S1.4 - Specification for Sound Level Meters.
S1.13 - Methods for the Measurement of Sound Pressure Levels.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D380 - Rubber Hose, Methods of Testing.
D622 - Rubber Hose for Automotive Air and Vacuum Brake System, Methods of Testing.
E11 - Wire-Cloth Sieves for Testing Purposes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The sprayer and duster unit shall consist essentially of a gasoline engine, air blower, solution pump, insecticide tank, nozzles, dust feeder, dust hopper, controls; and accessories, integrally assembled as a combination functional unit. The unit shall comply with all state and federal EPA requirements:

3.1.1 Standard commercial product. The sprayer shall, as a minimum, be in accordance with the requirements of this specification and shall be the

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manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the sprayer being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.2 First article. When specified (see 6.2), the contractor shall furnish one complete sprayer and duster unit for first article inspection and approval (see 4.2.1 and 6.4).

* 3.3 Materials. Materials used shall be, free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use, The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Safety. All rotating or reciprocating parts, and all parts subject to high operational temperatures, that are of such a nature or so located to be or become a hazard to the safety of the operating personnel shall be insulated, enclosed, or guarded to the extent necessary to eliminate the hazard. Each sprayer shall comply with OSHA regulations in effect at time of manufacture.

3.6 Performance. When operating with an air output velocity as indicated in table 1, as applicable to the equipment, the combined sound level of the engine exhaust, blower, and air discharge stream, will not exceed the recommended level as measured from the operator's station side of the sprayer and duster unit. When tested as specified in 4.4.1 through 4.4.4.4, the sprayer and duster unit shall be capable of dispersing insecticide formulations of oil or water based solutions, emulsions, and suspensions. The sprayer and duster unit shall also be capable of dispersing insecticide dusts and granules. The formulations and dusts shall be injected into the air stream, and homogeneously dispersed at a continuous steady rate. When the engine is operating at this continuous rated speed, the air nozzle shall discharge a steady stream of air at not less than one of the volume-velocity

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combinations given in table I. Liquid discharge rates shall be adjustable . between 30 and 100 gallons per hour (gph). Under windless conditions, the sprayer and duster unit shall be capable of dispersing water or oil based formulations, by means of the air output selected from table I, to a horizontal distance of 150 feet. At 100 feet from the nozzle, formulation particles of oil based insecticides shall have the characteristics given in table II. Dust and granule discharge rates shall be adjustable between 30 and 300 pound-force per hour (phr). Under windless conditions, the sprayer and duster unit shall be capable of dispersing dust formulations, having a fineness of not greater than 177 microns, to a horizontal distance of 150 feet. These requirements shall be met whether the equipment is in motion or stationary. There shall be no signs of damage after towing the type A sprayer and duster unit at 5 miles per hour (mph) over 4 by 4-inch timbers spaced approximately 3 feet apart on a paved surface; or after transport of the type B sprayer and duster unit, when centered over the driving axle of a flat bed truck, over the same or equal course. Towing or transport during spraying and dusting operations shall not jeopardize the operator nor interfere with his use of the controls.

TABLE I. Air output.

Air volume (cfm)	Air velocity (mph)
10,000	150
14,000	140
19,000	100

TABLE II. Oil based insecticides.

discharge rate (gph)	Distance from nozzle (feet)	Temperature of solution (not less than)	Ambient temperature (not less than)	Particle size (microns mass median diameter)
30	100	68° F	68° F	50 to 100
60	100	68° F	68° F	90 to 150

3.7 Design and construction. The equipment design and accessory installations shall provide accessibility for maintenance and service in the field. The design shall be such as to prevent conditions which may be hazardous to personnel or deleterious to equipment. The sprayer and duster unit shall be so constructed that all operating controls are within easy access to the operator when engaged in spraying or dusting. A comfortable seat, padded and covered with weather and spray resistant material, shall be provided, The seat shall be securely mounted in a manner to provide convenient control and safety under all operating conditions, including the engine starting system controls, engine speed control, air stream discharge

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direction control, rate of liquid insecticide ejection control, liquid formulation pressure control, and rates of dust and granule ejection control. Engine exhaust shall be directed away from the operator's station. Instruments and gages necessary for safety during spraying and dusting operations shall be within continuous view of the operator.

3.7.1 Type A. The type A sprayer and duster unit shall be mounted on, or integral with, a trailer. The trailer frame shall be of all steel construction with a minimum of splices and -joints. The trailer shall be provided with a single axle, designed for not less than the gross load imposed. Trailer wheels and bearings shall be the manufacturer's standard. The air nozzle shall be provided with an easily operated control, so that the air stream may be depressed not less than 110 degrees (°) from vertical on either side by rotating or tilting the air nozzle.

3.7.1.1 Wheels, tires, and tubes. Wheel hubs shall be fitted with ball or roller bearings of adequate size. Tires shall be tube or tubeless type, with highway tread. Wheels and tires shall be of rated capacity not less than the gross operating load imposed on each tire, measured at each wheel. Tires shall be of not less than 100 level quality. The manufacturer's current size of tires conforming to the specified requirements may be furnished. Rim rating shall conform to the TRA recommendations for the type and size of tires furnished. When tube type tires are furnished, inner tubes shall be heavy duty type, and shall be of proper size for the tires furnished. Tire flaps shall be provided in accordance with the TRA recommendations when required for the tube type tires provided.

3.7.1.2 Drawbar, landing leg, and hitch. A drawbar, with retractable landing leg, and flat plate hitch, or swivel type lunette ring hitch of 1-5/8-inch round bar steel stock and 3-inch diameter eye, shall be provided. The drawbar shall be an A-frame type, or straight steel bar or tube tongue type, bolted to the frame. When parked on level ground, with the landing leg in position, complete drainage of the tank shall be provided. The drawbar and landing leg and related parts shall withstand, without permanent deformation, the combined static and dynamic force due to towing, coupling, and uncoupling operations. The landing leg shall include a screwjack, shall operate freely, and shall be protected from malfunction caused by foreign matter. When in travel position, the landing leg shall remain locked.

307.1.3 Safety chains. Safety chains, conforming to SAE J697 or DoT Motor Carrier Safety Regulations, section 393.70(F), shall be provided and installed for connection of the drawbar to the towing vehicle.

3.7.2 Type B. The type B sprayer and duster unit shall be mounted on a structural steel skid base of welded construction. Each runner of the skid base shall have both ends upturned, and shall have a hole not less than 3. inches in diameter at each end to provide for handling by slings or chains. The base shall be of sufficient weight to insure stability in operation, sufficiently strong to keep the equipment in proper alignment when skidding the sprayer and duster unit along the ground, arid when raising or lowering.

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Tiedown devices shall be provided, as necessary, to insure safe operation when mounted on a flat bed truck. The air nozzle shall be provided with easily operated air stream direction controls, for 360° horizontal rotation, and not less than 110° depression on either side from the vertical. The air nozzle elevation control shall maintain any set position of the air nozzle, and shall not require unlocking for change of direction. Horizontal rotation of the air nozzle may be by powered worm gear drive or manually directed, and shall be designed to prevent horizontal rotation when the transporting vehicle is tilted 15° from horizontal in any plane. The positioning lock, if provided, shall be easily released from the operator's seat by foot or by hand.

3.7.3 Blower and air nozzle. Insecticidal liquids, dust, and granules shall be discharged through the air nozzle in a smooth stream, with a controllable trajectory. The blower or fan shall be fabricated of corrosion-resistant metal. Dusts and liquids shall be introduced into the high velocity portion of the air stream on the exhaust side of the blower or fan.

3.7.4 Dust and granule ejection system. The dust and granule ejection system shall include a hopper of not less than 3-1/2 cubic foot capacity, a mechanical dust and granule agitator or an air fluidizer, and an adjustable type erosion and corrosion resistant feeder control, designed to maintain continuous feeding of dry dust and granule insecticide, at rates between 30 and 300 phr. Construction and design of the hopper shall be such as to prevent escape of insecticide material. The hopper shall not tilt during change of elevation of the air nozzle. Dust channels shall be designed to prevent packing and accumulation, and to feed particles into the air stream at a point where the blast has a self-cleaning effect.

3.7.5 Liquid ejection system. The liquid ejection system shall include a liquid insecticide storage tank with a capacity of not less than 100 gallons for the type A unit, and 50 gallons for the type B unit, mechanical agitation, piping system with a positive displacement type pump, a replaceable basket type strainer, and controls. Valves, piping, and spray nozzles shall be designed to inject liquid into the air stream at even rates of flow between 30 gph and 100 gph as set. Hose connections shall be made at all points subject to vibration stress and where motion is required.

* 3.7.5.1 Insecticide storage tank. and. fittings. The tank shall be constructed of uncoated 0.0598 to 0.0747-inch thick stainless steel. Exposure to insecticide corrosion action testing solution (see 4.4.1.3) for not less than 100 hours, shall not cause the connecting hose and tank to flake, peel, blister, or dissolve and shall not develop leaks, electrolytic erosion, or metal transfer. The tank shall be provided with an accessible drain and shutoff valve. The tank bottom shall be sloped or rounded with a sump effect to promote complete drainage, and unrestricted insecticide circulation. Baffles shall be furnished, as necessary, to insure dynamic stability during spraying, dusting, and towing. The location of baffles shall not interfere with fluid agitation. The pump suction connection shall be extended into the tank; and provided with an intake of diameter and location to provide for

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insecticide flow, at a rate of 10 gallons per minutes (gpm), without fluid cavitation at any liquid level. Fittings and attachments integral with the tank, and not of the same metal as the tank, shall be compatible in the presence of insecticide. A filler opening, with minimum dimensions of not less than 10-1/2 by 14-7/8 inches oval or 13 inches square, and a quick clamping lid with tension locking device shall be provided:

3.7.5.2 Liquid agitator. The tank shall be provided with a mechanical corrosion-resistant agitator, designed to continuously and positively maintain currents over the entire area of the tank bottom. The agitator shall effectively mix all layers of insecticide solution or suspension.

* 3.7.5.3 Insecticide pump and piping system. The pump shall be a positive displacement piston type, with not less than two, ceramic or procelain-lined cylinders. Piston cups, O-rings, valve seats, nylon rollers, and wearing surfaces, as applicable, shall be capable of not less than 100 hours of pumping the insecticide test solution as specified in 4.4.1.3, without developing signs of deterioration, breakage, and imperfect seal. Pump valves and valve seats shall be of stainless steel, or shall consist of ground ceramic valves with replaceable nylon valve seats. Sealed bearings, when used, shall be unconditionally guaranteed by the manufacturer for not less than 2 years. Pressure lubricated bearings shall be protected from dust, dirt, and abrasive material. A nickel-copper strainer of not coarser than 30-mesh, with removable basket, shall be installed in the pump suction line. Pump replacement shall not require draining the tank. The piping system shall be designed for continuous circulation of insecticide, at a flow rate of 10 gpm, through the suction strainer, pump, and adjustable bypass pressure regulating valve, with return of unused fluid to the tank. A takeoff and flow control valve, with connection to the blower ejection nozzles, shall be provided. Circulation discharge return into the tank shall be designed to prevent entrapment of air. A pressure gage shall be installed in the high pressure piping or shall be incorporated in the regulating valve assembly. The pressure regulating valve shall be an unloading type, or shall be provided with a valve bypass for use during engine starting. Control and shutoff valve shall be of brass with ground metal seats, and shall have no internal rubber parts. Valves shall be of the type recommended by the manufacturer for the intended use. Gaskets and washers shall be smooth, seamless, compressible material unaffected by insecticides, Insecticide ejection nozzles shall be designed and installed to meet the performance requirements specified herein. Nozzles shall discharge into the high-velocity air stream on the exhaust side of the blower fan. Nozzles shall be equipped with a nickel-copper strainer of not coarser than 50-mesh.

3.7.5.4 Spray nozzles. The spray nozzles shall be designed and installed to meet the performance requirements specified herein. Provision shall be made for varying the discharge capacity and particle size by using interchangeable nozzles, nozzle tips, or adjustable nozzles. Nozzles shall be equipped with a 30-mesh pesticide resistant strainer. The strainers shall have sufficient surface area to minimize resistance to fluid flow.

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* 3.7.5.5 Hose. Hose shall be a commercial sprayer hose with the following characters tics:

Inner tube:	Smooth, seamless, insecticide-resistant material.
Outer tube:	Abrasion-, sunlight-, and weather-resistant black neoprene.
Reinforcing:	Two braids rayon cord.
Performance:	Working pressure: Manufacturer's rated pump output pressure plus 25 percent minimum burst: 4 times the working pressure.
	Change in outside diameter or length not over 6 percent when subjected to O to 800 pound-force per square inch (psi).
	Suction hose shall not collapse at a vacuum pressure of -28.3 feet of water (-24.0 inches mercury).

The inner tube shall not swell, flake, or disintegrate when subjected to the action of the petroleum distillates, chlorinated hydrocarbon solvents, insecticides, or aromatics.

3.8 Engine. The sprayer engine shall be a gasoline engine having horsepower, torque, and speed characteristics to meet satisfactorily all of the sprayer and duster unit performance requirements specified herein. Certification that the engine furnished is approved for this application shall be available during fabrication. The engine shall be complete with the accessories normally furnished with the sprayer and duster unit, including not less than the following:

a. A 12 volt electrical system, including a starter, battery charging alternator (shielded from insecticide and weather) of not less than 15-ampere rating, voltage regulator, ammeter, and a 15 ampere dry charged battery without a filling of electrolyte.
b. An engine housing.
c. A fuel tank of sufficient capacity for not less than 3 hours of continuous operation at rated speed.
d. A fuel filter.
e. A speed governing system, preset but adjustable to maintain the manufacturer's recommended continuous operating speed.
f. An engine hour meter, when specified (see 6.2).
g. An air cleaner.

The engine shall meet the requirements specified herein using gasoline that is unleaded or having a low-lead content. Low-lead gasoline shall not contain more than 0.5 gram of tetraethyl lead per gallon of gasoline.

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3.8.1 Noise level. Unless otherwise specified (see 4.4.4.6 and 6.2), equipment noise level shall conform to current industry standards except that the noise exposure for 8 hours shall not exceed 85 decibels at the operator's position when measured in accordance with ANSI S1.13 on the A scale of a sound level meter conforming to ANSI S1.4.

3.8.2 Electromagnetic interference characteristics. When specified (see 6.2), the equipment shall conform to the electromagnetic characteristic requirements and test limits for class no. IIID equipment, as specified in MIL-STD-461.

3.8.3 Fungus resistance. When specified (see 6.2), electrical connections, including terminal and circuit connections, components, and circuit elements, shall be coated with varnish conforming to MIL-V-173 except that:

a. Components and elements inherently inert to fungi or hermetically sealed need not be treated.
b. Components and elements whose operation will be adversely affected by the application of varnish shall not be treated.

3.9 Toolbox. The sprayer and duster unit shall be provided with a metal. toolbox. The toolbox shall have a hinged lid with a locking device that will keep the lid closed when subjected to vibration and rough travel. The toolbox shall be accessible and securely fastened to the equipment in a protected area. The toolbox shall have ample capacity for all necessary field operation and maintenance tools and instructions.

* 3.10 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 roils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

* 3.11 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

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3.12 Identification plate. An identification plate will be furnished by the contracting officer for each sprayer and duster unit. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each sprayer and duster unit in a conspicuous place with brass screws or bolts not less than 1/8-inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.13 Instruction. plates. The sprayer and duster unit shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.14 Workmanship. Workmanship shall be the best quality used in industry for sprayer and duster units. Defective components, parts, and assemblies, which have been repaired or modified to overcome deficiencies shall not be furnished. Bolted and riveted construction, when utilized, shall be in accordance with the highest standards of the industry.

3.14.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel. to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.14.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.14.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.14.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.14.5 Castings. All castings shall be sound and free from patching,

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misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein, Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one sprayer and duster unit when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3, the tests of 4.4, and, when specified, the preproduction pack inspection of 4.5 (see 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4.3, and the packaging inspection of 4.5.

4.3 Examination. Each sprayer and duster unit shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or-lessening maximum efficiency shall constitute cause for rejection.

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4.4 Tests.

4.4.1 Testing liquid and. solutions. Testing liquids and solutions, in accordance with 4.4.1.1 through 4.4.1.3, shall be used during testing for performance and water based insecticides and oil based insecticides, and for resistance to insecticidal corrosion action, as applicable.

4.4.1.1 Water based insecticides. The test solution to be-used for droplet distribution testing (see 4.4.4.1) of water based insecticide shall be salt saturated water.

4.4.1.2 Oil based insecticides. The test liquid to be used for droplet distribution testing (see 4.4.4.2) of oil based insecticides shall be number 2 diesel oil.

4.4.1.3 Corrosive action resistance test solution. The test solution to be used for testing resistance to the corrosive action of insecticide shall be an emulsion prepared by mixing 1/2 gallon of 60 percent Naled (dibrom) emulsifiable concentrate with 49-1/2 gallons of water (unit quantity - 50 gallons gross).

4.4.2 Testing dust and granule mixture. A commercial insecticide powder, devoid of lumps, and of such fineness that not less than 99 percent passes a U.S. standard number 80 in accordance with ASTM E11 (177 micron) sieve (dry test), shall be used as the insecticide dust and granule mixture for test purposes.

4.4.3 Production unit functional operation tests. After assembly, each production sprayer and duster unit shall be serviced and fueled for functional testing. The insecticide storage tank and dust bin shall be filled with the testing liquids, solutions, and dust applicable to the intended test procedures. Type B sprayer and duster unit shall be securely mounted at center point over the axle of a flat bed truck trailer, or other suitable transporting medium. Test liquids, dust and granules, and fuel remaining at the conclusion of all tests shall be removed and discarded.

4.4.3.1 Roadability. and transportability tests. Prior to conducting any other test, the sprayer and duster unit shall be tested for readability and transportability with a capacity load of solution, dust, or granules. The unit shall be moved so that the wheels pass over not less than three 4- by 4-inch timbers both alternately and simultaneously. The timbers shall be spaced 3 to 4 feet apart. The test shall be run at not less than 5 mph and shall be repeated three times. The type B sprayer and duster unit shall be dragged not less than 10 feet by means of a cable attached at one corner of the skid base. The type B units shall be attached to a flat bed truck or other medium for the readability test. The sprayer and duster unit shall be mounted directly over the transporting vehicle's axle for the roadability test. Evidence of damage which might affect subsequent tests shall be cause for rejection of the unit and discontinuance of the tests.

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4.4.3.2 Air stream volume-velocity output test. The engine speed shall be set at the manufacturer's recommended continuous operating speed. The velocity of the air at the outlet shall be measured at not less than five equidistance locations from the top to the lower edge of the air nozzle outlet, averaged, and the volume-velocity combination shall be determined. The test shall be continued for 1/2 hour. Readings shall be taken every 5 minutes and averaged to determine conformance to 3.6, table I.

4.4.3.3 Insecticidee discharge flow rate tests. With air output at any of the volume-velocity combinations specified in 3.6, table I, the liquid discharge, and the dust and granule discharge, shall be measured for a sufficient period of time to determine the capacity of the sprayer and duster unit to disperse liquid at the 30 and 100 gph rates, and dust and granules at the 30 and 300 phr rates. The liquid discharge test shall be performed either with the water based (see 4.4.1.1), or the oil based (see 4.4.1.2), insecticide test liquids.

4.4.4 Field operation tests. The first article shall be operated not less than 10 minutes spraying, and not less 10 minutes dusting. All controls shall be operated to determine proper function. With the engine operating at the recommended continuous operating speed, the trajectory and reach of the air stream, in all directions applicable, to the designed range, shall be observed. Uniformity of spray pattern shall be confirmed. The safety of the operator, during spraying and dusting, at a travel speed not less than 5 mph shall be noted at all times.

4.4.4.1 Insecticide droplet distribution. test. Droplet distribution at a range of 150 feet shall be tested, using the water based (see 4.4.1.1), testing solution. During the test, the first article shall be stationed at the proper range, shall have the trajectory of the test solution bearing air stream shielded from the wind, and shall be continuously and steadily discharging at one of the volume-velocity combinations specified in 3.6, table I. The test solution shall be injected into the air stream for a period not longer than 30 seconds at each flow rate of 30 gph and 100 gph. Five 1- by 3-inch clean glass slides, for each flow rate, shall be exposed to the spray at the 150-foot range. Effectiveness of the droplet distribution shall be determined by recovery of salt from the slides, after evaporation of the droplets. Failure to recover salt on the slides at the 150-foot distance shall be cause for rejection.

4.4.4.2 Insecticide droplet article size test. Droplet particle size at a range of 100 feet shall be tested, using the oil based (see 4.4.1.2), testing solution. During the test, the first article shall be stationed at the proper range, shall have the trajectory of the test solution bearing air stream shielded from the wind, and shall be continuously and steadily discharging one of the volume-velocity combinations specified in 3.6, table I. The test solution shall be injected into the air stream for a period not longer than 30. seconds for each particle size test. Tests shall be conducted at both flow rates given in 3.6, table II. Twenty, 1- by 3-inch glass slides, treated with cleaning solution and coated with carbon black, or an oleophobic

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substance which will cause oil droplets to retain some of their convexity and prevent irregular spreading on the slide, shall be prepared. One set of 10 slides shall be used for the 30 gph particle size test, and the second set for the 60 gph test. For each test, the slides shall be positioned 100 feet from the sprayer and duster unit. After exposure, the slides shall be examined with a 50-power microscope with eyepiece micrometer and mechanical stage. At least 200 particles on each slide shall be measured. After applying a spread factor of 0.33, failure of the majority particle size to be within the range - specified in 3.6, table II, for the discharge rate, shall be cause for rejection.

4.4.4.3 Dust and granule distribution test. Dust and granule distribution tests shall be performed at a range of 150 feet, using a testing dust and granule mixture specified in 4.4.2, of a density to cause falling to the ground as air speed diminishes During the tests, the first article shall be stationed at the proper range, shall have the trajectory of the insecticide bearing air stream shielded from the wind, and shall be continuously and steadily discharging at any of the volume-velocity combinations specified in 3.6, table I. Independent tests shall be performed at the 30 to 300 phr ejection rates. Insecticide dust and granules shall be ejected into the air stream for a period not longer than 30 seconds, for each test distributions Dust and granules shall be collected on a 10 foot square canvas with smooth plastic finish of a color contrasting to the insecticide test dust and granules, placed on an open flat surface. Test dust recovered from the canvas after each test shall be in sufficient quantity to be measurable in terms of pounds per acre.

4.4.4.4 Insecticide resistance test. Not less than 50 gallons of insecticide, composed as specified in 4.4.1.3, shall be placed in the tank of the first article. Under normal operating conditions, the insecticide shall be recirculated without discharge and refilling, for not less than 4 hours daily, for 5 days, without removing the test solution between daily runs. On completion of the test, the tank interior surface or coating, tank bottom, connecting hose, the suction strainer basket, gaskets, and all accessible surfaces which were continually exposed to the insecticide solutions shall be examined for flake, peel, blister, and dissolution. The tank and piping system shall be examined for any sign of leakage. The insecticide storage tank shall be completely drained, and shall be examined for signs of stoppage, sediment accumulation, stagnation traps, rust, and air lock conditions Pump cylinders, valves, piston cups, and O-rings shall be disassembled and examined for ero- sion, pitting, electrolytic metal transfer, deteriorations breakage, gasket damage, and abnormal wear. Precharged air chambers, when used, shall be tested to determine retention of the original pressure. Evidence of chemical damage, erosion, electrolytic metal transfer, and distorted parts shall be cause for rejection.

* 4.4.4.5 Connecting hose test. A connecting hose or hose assemby, which has been used during testing with all three test liquids (see 4.4.1 through 4.4.1.3), shall be removed for testing. The hose assembly shall be tested in accordance with the straight burst, outside diameter expansion and elongation

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and contraction test of ASTM method D380, and vacuum test of ASTM method D622, and compared after testing to the characteristics before testing, to determine conformance to 3.7.5.5. Failure to conform to the characteristics specified, signs of flaking, disintegration, or swelling due to chemical action of the testing liquids, and collapse or distortion to the point of flow restriction during vacuum testing, shall be cause for rejection and replacement of the hose with new hose which has been tested and accepted.

4.4.4.6 Noise level test. Compliance with the noise level limits of 3.8.1 shall be verified by measurement. Noise levels shall be measured using a sound level meter in accordance with ANSI S1.4, type 1. The meter shall be set for the A-weighting "slow" mode of operation. The noise level test shall be set up and conducted in accordance with the guidelines specified in ANSI S1.13 for the "Field" measurement method.

4.4.5 Test for electromagnetic interference. characteristics. When electromagnetic interference characteristic limits are required (see 3.8.2), the characteristics of the sprayer and duster unit shall be tested in accordance with MIL-STD-462.

4.5 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-S-12651. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

5. PACKAGING

* 5.1 Preservation. packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-S-12651 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

6.1 Intended use. The high velocity air stream distribution of liquid, dust, and granular pesticides is used for control of soil and vegetation pests, and for the control of disease vectors, where application of pesticides by other methods would be time consuming or impracticable.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of sprayer and duster unit required (see 1.2).
c. Specifications and standards (see 2.1.1).
d. When first article is required (see 3.2, 4.2.1, and 6.4).
e. When engine hour meter is required (see 3.8).
f. When noise level is other than specified (see 3.8.1).
g. When electromagnetic characteristic limits are required (see 3.8.2).
h. When fungus resistance treatment is required (see 3.8.3).
i. Color of finish coat required (see 3.10).

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j. When lubrication required is other than the manufacturer's standard practice (see 3.11).
k. When a preproduction pack inspection is required and time frame required for submission (see 4.5).
l, Level of preservation and level of packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

* 6.4 First article. When a first article inspection is required (see 3.2 and 6.2), the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one sprayer and duster unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-23923D	SPRAYER AND DUSTER UNITS, INSECTICIDE, MIST-DUST BLOWER TYPE: GE 30 TO 100 GAL/H LIQUID, 30 TO 300 LB/H DUST (COMMERCIAL)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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